

FLEX-UF: Supplementary Methods and Controlled Experiments

Anonymous CVPR submission

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S1. Scope and Reproduction

This supplement records CPU analyses of the frozen evaluation archive and a prospective protocol for the ongoing depth study. The main manuscript uses the recorded mixed reconstructions of the shared-latent e15 codec. The new control-path and spatial-assignment analyses reuse its stored source-error tables. They produce candidate decisions for later replay, without executing the codec or measuring new inference times. The independent D2/D4/D6 training continues separately; no intermediate checkpoint is treated as a completed depth experiment.

The stand-alone paper repository contains the plotted source tables, source hash manifests, editable figure scripts and inspection results under `data/refresh20260927`. Figures can be regenerated on CPU using `scripts/build_figures.py` and `scripts/build_extended_figures.py`. The parent research repository also retains the scripts that derive these summaries from the raw records and checkpoints. Hashes identify the files inspected during this revision; they do not retrospectively establish the exact executable used by a historical run whose code hash was not recorded.

S2. Tracing the Reconstruction Reference

Two measurement paths need to be distinguished. The source-table producer loads the e15 network and a separate released-weight reference. It computes the candidate tile errors $M_{t,k}$ from e15 and the scalar reference error R from the released synthesis network, both on the padded source. The final-reconstruction evaluator instead loads e15 alone, calls its full-frame synthesis path, crops both full and mixed outputs, and measures their difference in source quality. Its raw field named `psnr_release` is consequently a fine-tuned full-depth reference in the inspected code (Fig. S2).

A CPU comparison of the two state dictionaries finds 255 shared encoder/entropy tensors with no changes. All 143 inherited synthesis tensors that can be compared have changed, with a maximum absolute parameter difference of 0.2481. Thus the reporting path cannot be identified with released synthesis merely because it executes all twelve trunk

blocks. After exact key remapping, all 398 inherited warm-start tensors also match the archived official D12 release. Extra adapter/router tensors are outside this comparison. This second check ties the inspected warm start to the released parameters without assuming that its filename proves provenance.

This distinction limits the interpretation of threshold crossings. The recorded 7/263 RGB and 49/263 YUV losses above a nominal 0.1 dB target combine differences in reference, image support, metric and the influence of the shared repair/head. They do not identify the contribution of any single factor. The paired MAC accounting remains attached to the same recorded maps, but the exact quality gap to the cropped released decoder cannot be recovered from the exported JSON alone.

The required replay should retain both full-depth anchors, the padded and valid-pixel errors, and the final mixed reconstruction. Selection and final verification must use the same declared reference and metric. A failed plan should trigger an explicit fallback whose bytes and runtime are recorded.

S3. What Additional Synthesis Depth Buys

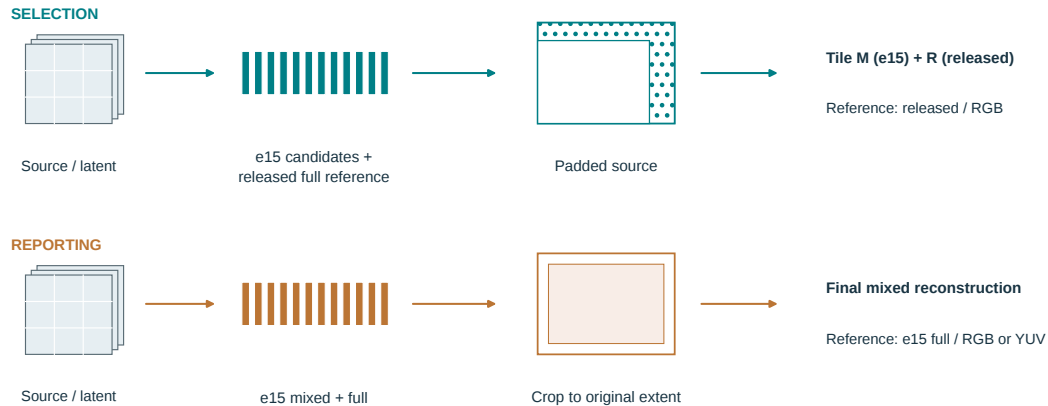
Each column of the archived source-error table was produced by decoding a uniform-depth map and measuring per-tile source MSE. Its equal-area tile average is therefore the padded RGB MSE of that actual uniform reconstruction. Comparing this mean to the separately evaluated released reference gives mean losses of 0.2045, 0.0968, 0.0605 and 0.0379 dB for 6/8/10/12 blocks over all 265 pairs. The corresponding modelled MAC savings are 39.12%, 24.22%, 13.03% and -0.95% . The negative all-deep saving reflects shared repair overhead relative to the full-frame reference. These values use padded support and cannot substitute for cropped released-reference replay.

For adjacent depths, the local quality improvement is $g_t = 10 \log_{10}(M_{t,k}/M_{t,k+1})$. The frame-median improvement, averaged across pairs, is 0.0993 dB for 6-to-8 blocks, 0.0320 dB for 8-to-10, and 0.0213 dB for 10-to-12. The respective mean within-frame interquartile ranges are 0.0748, 0.0340 and 0.0238 dB. This quantifies both diminishing returns and variation within the same image. The average



Figure S1. **Nominal targets and final reporting answer different questions.** **a**, Recorded average cropped RGB loss against MAC saving. **b**, Router RGB and YUV 6:1:1 loss distributions at a nominal 0.1 dB target. **c**, Counts above that numerical target, among the 263 common cases and with 10^{-4} dB tolerance. Selection uses a released reference on padded RGB; reporting uses cropped e15 full-frame synthesis. The counts therefore do not isolate a common-reference guarantee failure.

a Selection and reporting use different reconstruction anchors and image support



CPU audit: 255 shared encoder/entropy tensors unchanged; 143/143 inherited synthesis tensors changed.

Required comparison: reconstruct both anchors, use the same valid pixels and metric, then verify the selected output.

Figure S2. **Reference and support are part of the measurement.** Candidate tile errors come from e15, whereas the table denominator comes from a separate released-weight reference. Final reporting uses the cropped full-depth e15 reconstruction. The CPU tensor comparison establishes a weight difference in the inspected checkpoints. The figure traces code paths and does not depict newly decoded images.

080 fractions of tiles whose error increases are 4.37%, 11.55%
081 and 11.41%; deeper synthesis is not uniformly better at every
082 location.

Figure S3 uses equal frame-QP weighting, rather than
letting larger images dominate through their tile counts.
Sequence-cluster intervals retain all five QPs together. Each

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tile comparison still changes the depth of its neighbours because the underlying maps are uniform. It is a conditional reconstruction diagnostic, not a causal one-tile intervention or a completed independent-codec depth ablation.

S4. Auditing the Scalar Control Search

For router scores $\ell_{t,k}$ and costs c_k , increasing the price β in $k_t(\beta) = \arg \max_k (\ell_{t,k} - \beta c_k)$ cannot increase selected cost. To see this, write the optimality inequalities at $\beta_1 < \beta_2$ for their respective choices k_1, k_2 and add them. The result is

$$(\beta_2 - \beta_1)(c_{k_1} - c_{k_2}) \geq 0. \quad (S1)$$

No corresponding statement follows for source error $M_{t,k}$, because that quantity is absent from the score being optimised. A feasibility test at the most expensive endpoint can therefore miss a feasible middle assignment, and distortion-based bisection can miss a later feasible interval.

We enumerate pairwise score-line intersections, representatives of the open intervals between them, and the endpoints of the archived range $[-20000, 20000]$. Archived controls are included explicitly to retain their floating-point tie behaviour. Identical maps are merged. Among feasible maps, the selected candidate maximises modelled MAC saving, with table loss as a tie-breaker. This is a finite-path audit in float64, rather than an optimiser over all possible tile assignments.

The cost is monotone on every inspected router path, while source loss has at least one reversal on 227/265 paths. At 0.05 dB, enumeration improves 3 of 204 previously feasible plans, by 0.1626 percentage points on average, and recovers one additional feasible case. At 0.1 dB it improves 2 of 263 plans, by 0.0277 points on average. No paired saving changes at the four larger targets. The largest missed saving is 30.45 points at 0.05 dB, so a small mean difference should not be read as a uniform per-frame bound.

We strengthen the blind baseline in the same way. For ordered dithering, all changes occur at a finite set of Bayer-rank thresholds and rung boundaries. Enumerating those maps reveals source-loss reversals on 232/265 paths. It improves 3/204 paired plans at 0.05 dB and 1/263 at 0.1 dB, with mean gains of 0.0192 and 0.0028 points, respectively. Six additional cases become feasible at 0.05 dB. The quality of these newly selected maps has not been measured after final mixed reconstruction, so they do not replace the main manuscript’s reconstruction curves.

The audit reproduces every archived feasible control and cost, verifies cost monotonicity, and checks that enumeration never discards a better archived feasible saving. An independent three-exit example verifies the non-monotone failure mechanism against a dense control grid.

S5. Separating the Histogram from Spatial Placement

A map can save arithmetic by selecting many cheap exits even if their locations contain no useful information. To isolate placement, let $p_k = N^{-1} \sum_t \mathbf{1}[k_t = k]$ be its fixed exit histogram and let π be a uniformly random permutation of its assignments. Each tile has marginal exit probability p_k , giving the exact expectation

$$\mathbb{E}_\pi \left[\frac{1}{N} \sum_t M_{t,k_{\pi(t)}} \right] = \frac{1}{N} \sum_t \sum_k p_k M_{t,k}. \quad (S2)$$

We report $G = 10 \log_{10}(\mathbb{E}_\pi[\overline{M}_\pi]/\overline{M})$, where $\overline{M}$ is the recorded map’s table MSE. Positive G indicates useful spatial placement at exactly the same histogram and MAC cost. The logarithm is outside the expectation: this is not expected PSNR over random reconstructions.

At 0.1 dB, source-informed search has a mean placement gain of 0.0321 dB, the router 0.0213 dB, and dithering 0.0007 dB over their respective shuffled histograms. The router’s 95% sequence-cluster interval is [0.0167, 0.0262] dB; dithering’s is [-0.0002, 0.0017] dB. Uniform-depth maps have zero gain by construction. Among the 263 paired router maps, 204 have positive gain, seven have negative gain and 52 use one exit only. This diagnoses content association conditional on a source-calibrated histogram; it does not validate held-out autonomous calibration.

For each map we also solve the exact fixed-histogram assignment problem on M . Repeating each exit column according to its count produces a square linear assignment problem. Its minimum table MSE gives a conditional upper bound on placement value. The router’s mean regret to that optimum is 0.0044 dB at the 0.1 dB target. Every source-search map attains its own fixed-histogram optimum, as expected for a separable Lagrangian minimiser. The optimum still requires the complete source table and carries the same information-acquisition cost. Replaying shuffled and optimal maps through the mixed decoder is the next required ablation.

S6. A Common Cap on Delivered RGB Loss

The stored mixed reconstructions permit a second comparison under a common reporting anchor. For each frame-QP pair and policy, we collect its up to six archived nominal-budget maps, deduplicate identical maps, and retain only candidates whose measured cropped RGB loss meets the chosen cap, with tolerance 10^{-4} dB. The evaluated full-frame e15 reconstruction is an explicit fallback with zero relative loss and zero MAC saving. The cheapest eligible candidate is selected; ties prefer lower loss.

This keeps all 265 pairs at every cap. At 0.1 dB, the retained maps save 29.19% of modelled decoder MACs

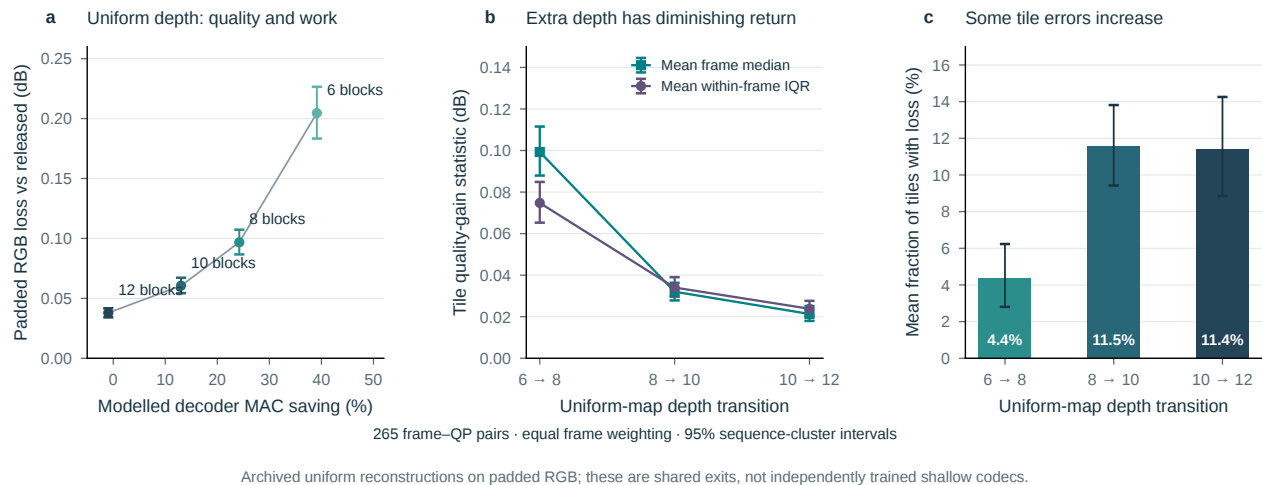


Figure S3. **The value of additional depth varies across regions.** **a**, Uniform-depth padded RGB quality against the released anchor and the recorded MAC model. **b**, Mean frame-median tile improvement and mean within-frame IQR for adjacent depths. **c**, Mean fraction of tiles whose error increases. Error bars are 95% sequence-cluster intervals. These are actual archived uniform reconstructions; neighbouring depths change together, and independent shallow codecs are not represented.

for source-informed search, 27.75% for the router, 24.81% for dithering and 22.98% for uniform depth. The router-dither difference is 2.93 percentage points, with a paired 95% sequence-cluster interval of [2.11,3.79]. Their mean achieved RGB losses are 0.0786 and 0.0791 dB, respectively. Full-frame fallback is selected for one router case, two search cases, two dither cases and eleven uniform cases. At a 0.3 dB cap, the routing margin falls to 0.35 points [0.11,0.64] (Fig. S5).

Every selected output was measured in the original archive, and no new image or interpolated operating point is introduced. This diagnostic is nevertheless retrospective and source-aware: it selects among candidates after their final quality is known. It neither accounts for the cost of obtaining those candidates nor establishes an autonomous policy. Its finite candidate pools are not exhaustive, and its reference is fine-tuned e15, not the released model. Within those boundaries, the observed routing margin survives both a common delivered-loss cap and explicit fallbacks.

A descriptive influence check removes each sequence with all five QPs. At 0.1 dB, the remaining mean router-dither margin ranges from 2.80 to 3.12 points; this sensitivity range is not a confidence interval. Each QP also has a positive mean margin. Removing the three pairs where either method falls back leaves 2.91 points over 262 pairs, although this restricted subset changes the estimand. At 0.5 dB, 262 pairs tie and all three positive differences belong to one sequence. These checks are reproducible from cap_influence_audit.json.

S7. Shared-Exit Training Provenance

The released warm start, shared-exit fine-tuning and independent depth trainings are three distinct checkpoints/protocols. The archived shared-exit metadata lists 379,614 training images, batch size eight and 16 configured fine-tuning epochs. The inspected launcher selects 512-pixel crops through the late part of the upstream schedule, freezes the encoder/entropy path, and enables mixed-tile training, depth-scaled adapters, grid repair, adjacent feature distillation and a full-depth anchor. Its anchor and distillation coefficients are 10 and 1, respectively; the anchor code scales its batch-mean reconstruction error by the batch-mean λ unless the separate per-sample option is enabled. The e15 checkpoint stores weights, epoch and architectural configuration, not the complete launch command. Thus the launcher and metadata document the inspected intended setup, rather than proving every historical process argument.

The evaluated router checkpoint explicitly stores live inputs stem, qp, soft-label cross-entropy, temperature ratio 0.1, training cost weight 1.3×10^{-5} , 48 stem projection channels and 256 hidden units. It has no spatial-context convolution. General-purpose latent/scale/bit pathways remain in the class, but do not provide variable content to this checkpoint. The decoder feature is pooled by tile mean and standard deviation, and the QP is divided by 63. Checkpoint inspection is recorded in design_audit.json. This corrects the broader description one would obtain by reading the router class defaults alone.

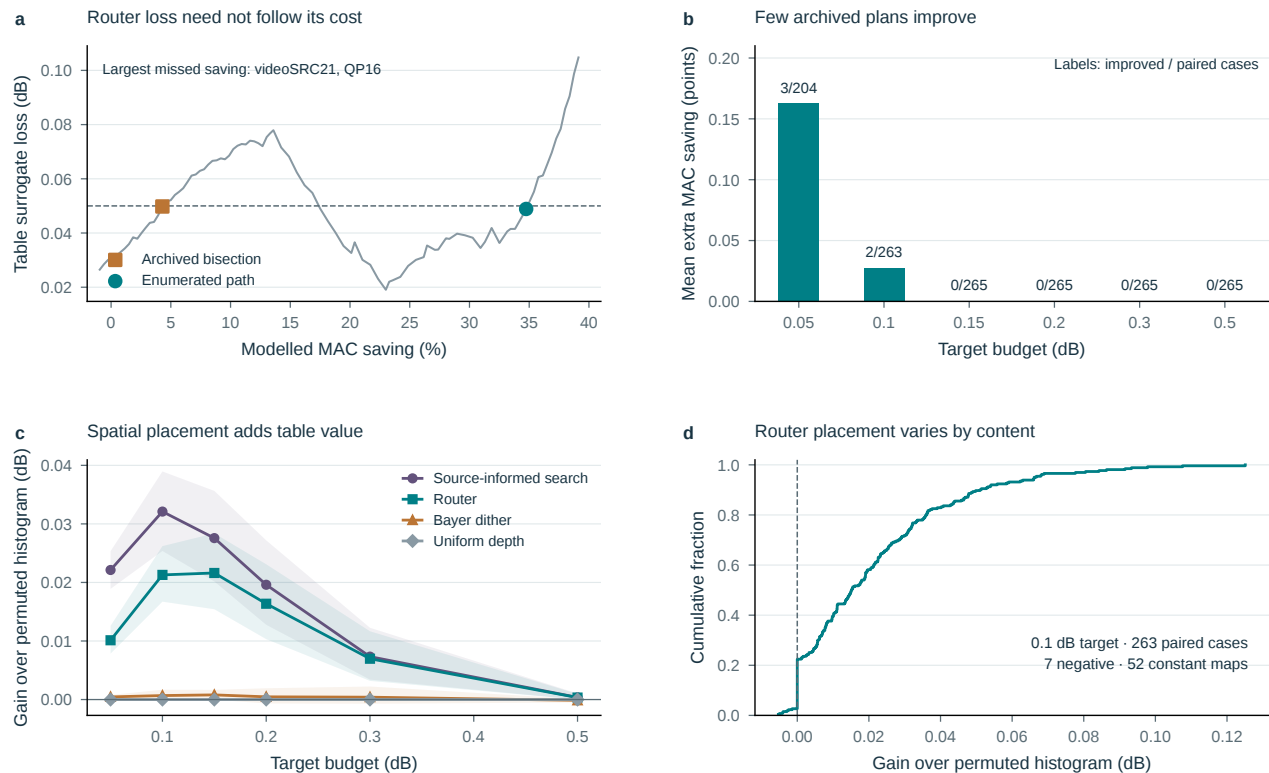


Figure S4. **Control-path and assignment diagnostics.** **a**, The explicitly selected largest missed router saving illustrates non-monotone table loss. **b**, Mean paired improvement from enumerating the router path; labels show improved and paired counts. Newly feasible cases are excluded from these means. **c**, Exact expected-shuffle comparison at unchanged exit histograms, with 95% sequence-cluster intervals. **d**, Router placement-gain distribution at 0.1 dB. All results are CPU analyses of padded error tables; none measures new final-image quality or latency.

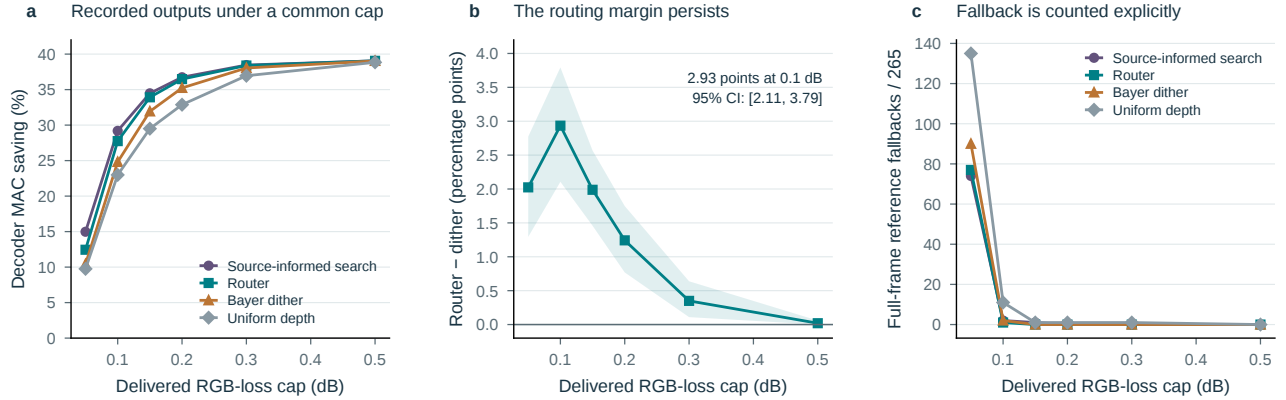
The companion run record matches the checkpoint path, codec epoch, live inputs and stored evaluation statistics. It records 6,000 updates, batch size six, 512-pixel crops, seed zero, initial learning rate 10^{-3} and a regret coefficient of one. The inspected trainer uses AdamW with weight decay 10^{-4} , cosine scheduling and gradient clipping at norm one on Open Images. Its online agreement monitor withholds tiles within training images; the final label-agreement pass draws fresh images from the same shuffled epoch. Neither statistic is used as the manuscript’s reconstruction or held-out calibration result. The stored head contains 148,176 parameters, including its inactive input branches.

S8. The Independent Depth Study

The ongoing experiment retains the first two, four or six synthesis trunk blocks of the DCVC-UF intra architecture and trains each complete codec under the pinned official recipe. The encoder and entropy weights learn alongside the decoder. Each model consequently has its own representa-

tion, distinct from the shared-latent exits evaluated in the main paper. Figure S6 shows a possible later deployment path, not an implemented six-expert result.

The pinned training protocol. The source is Microsoft’s image-training recipe at commit `cbdae87a5445` [4]. All 384,795 images in Open Images train_0/1/2 are retained as lossless decoded-RGB PNGs, with pixel equality checked during preparation. Each independent model uses seed 42, Xavier initialisation, batch size 16, AdamW with the upstream defaults, and gradient clipping at norm 0.1. QP is sampled from 0–63 with logarithmic λ interpolation from 10 to 2048. The original loss, including its literal 0.0833 coefficient, and the upstream crop, flip and colour transforms are retained. Table S1 records the full 105-epoch schedule. There are 24,049 batches per epoch and 2,525,145 scheduled updates per model. With the official shuffled `drop_last=True`, eleven images per epoch do not enter a batch; no image is permanently excluded from the training



265 frame-QP pairs throughout · fine-tuned e15 anchor · retrospective source-aware candidate selection

Figure S5. **Controlling delivered loss within the recorded candidate pool.** **a**, Maximum saving among actually measured eligible outputs, including a full-frame e15 fallback. **b**, Paired router–dither difference with 95% sequence-cluster intervals. **c**, Explicit fallback counts. All 265 pairs are retained, with RGB-loss tolerance 10^{-4} dB. This retrospective source-aware comparison excludes the cost of candidate generation and makes no released-reference or latency claim.

Table S1. **Unshortened official image-training schedule.** Epoch indices are zero-based. Each row applies to D2, D4 and D6; the architectural intervention is the retained synthesis depth.

Epochs	RGB crop	Learning rate
0–44	256^2	2×10^{-4}
45–69	256^2	5×10^{-5}
70–89	256^2	10^{-5}
90–94	512^2	2×10^{-4}
95–98	512^2	5×10^{-5}
99–102	512^2	10^{-5}
103–104	512^2	10^{-6}

manifest.

Training uses FP32 with the upstream TF32 flags and `torch.compile`, with `fallback_random=True` to retain the PyTorch random-number path. AMP is disabled. The instrumented eager step was checked against the upstream step at identical inputs and RNG state. Compilation can change floating-point operation order, so protocol parity does not imply bitwise-identical training trajectories. Check-point manifests retain the execution mode and source hashes.

Isolating depth. Released D12 is a useful deployment anchor, but a D12 trained with the same data manifest, 105-epoch schedule, optimisation and initialisation protocol is necessary to isolate depth from training history. D8 and D10 should be prioritised after the initial depth frontier and

this control are available. The main runs retain the official recipe; distillation and prefix initialisation are separate variants rather than silent training changes.

Isolating routing. The evaluation should progress from full-frame D12 to patchwise D12, the best validation-selected fixed expert, blind mixtures, simple image statistics, the MLP and source-informed search. Full-frame versus patchwise D12 measures the cost of tiling and entropy resets before routing receives credit for any gain. A histogram-preserving shuffle separates the chosen expert proportions from their spatial placement. A model below the dataset-mean frontier may still help a subset of patches. Expert removal should therefore be evaluated by leave-one-expert-out routing at matched constraints, not justified by mean full-image quality alone.

A concrete router hypothesis. A lightweight encoder-side predictor can estimate each expert’s distortion and rate relative to a cheap reference. Let $z_{t,k}$ indicate the expert assigned to patch t and $n_k = \sum_t z_{t,k}$. For measured grouped-runtime tables $\tau_k(n_k)$, a proposed system objective is

$$\hat{J} = \sum_{t,k} z_{t,k} (\hat{D}_{t,k} + \lambda \hat{R}_{t,k}) + \mu \sum_k \tau_k(n_k). \quad (\text{S3})$$

Routing, packing and scatter/gather work must also enter the measured total; the expression isolates the expert-runtime dependence. An independent per-patch latency label approximates this coupled problem. A bounded post-

processing variant could move a patch from k to j according to its predicted RD change and the measured marginal time $\tau_k(n_k - 1) - \tau_k(n_k) + \tau_j(n_j + 1) - \tau_j(n_j)$. The number of passes and decision-time limit should be fixed on validation images. Any improvement must include this extra control work.

Regression or regret-weighted training should be compared against ordinary exit-label cross-entropy. A second-stage top-two verification can be restricted to small predicted margins, with its threshold selected on held-out calibration images and its extra encoding work included. Neither a large prediction margin nor a low average regression error establishes a framewise quality guarantee. These are prospective experiments, not measured components of the current shared-exit system.

Spatial competition between complete codecs already provides a close precedent [1]; adaptive compression and patch early exit provide further controls [2, 5, 7]. The testable direction here is depth-heterogeneous selection with measured rate-distortion-time costs and reduced encoder search. Practical codec studies motivate including device execution and perceptual assessment [3, 6].

S9. Evaluation Contract for the Next Stage

Expert training, router training, calibration and final evaluation should be split at image level. Patches from the same image must stay in one split. The four fixed validation crops in the current training monitor are health checks; a broader intermediate evaluation should use the available 100 DIV2K validation images, followed by the declared final Kodak/CTC/CLIC scope. This expanded evaluation is proposed and has not been completed.

Each codec needs a true-byte rate-distortion curve. Equal QP does not imply equal rate across independently learned codecs. Comparisons should use their common measured support, reporting infeasible targets and avoiding extrapolation. Rate includes main and hyperprior payloads, model identity, quality/control parameters and framing headers, divided by original pixels. Quality is measured after cropping valid pixels; the exact RGB/YUV formula and perceptual metrics are declared before test-set selection.

Timing requires the same GPU, precision, kernel path and warmup, with alternated paired execution order and retained raw trials. Encoder, decoder and complete round-trip times are separate outcomes. CUDA-event module time and end-to-end wall time must remain separately labelled. Grouping, scatter/gather, entropy work, stream parsing, transfers and fallback are included where they occur. The full codec bank's resident weight memory and peak activation memory are both reported.

The primary success criterion is a positive net runtime advantage over the best validation-selected blind control on common achieved rate-distortion support. Critical compar-

isons should then be repeated across training seeds. A content bootstrap at fixed weights measures a different source of variation and cannot replace those repeats.

S10. Analytical Scenarios for the Pending Depth Measurements

The independent depth study has no completed RD or latency frontier yet. To make the design hypothesis inspectable, Fig. S7 separates exact architectural parameter counts from a hypothetical runtime model. For retained depth d , let f be the assumed fraction of full-depth time spent in the twelve-block trunk and let h be added routing/grouping overhead normalised by the original full-depth time. Under linear trunk-time scaling,

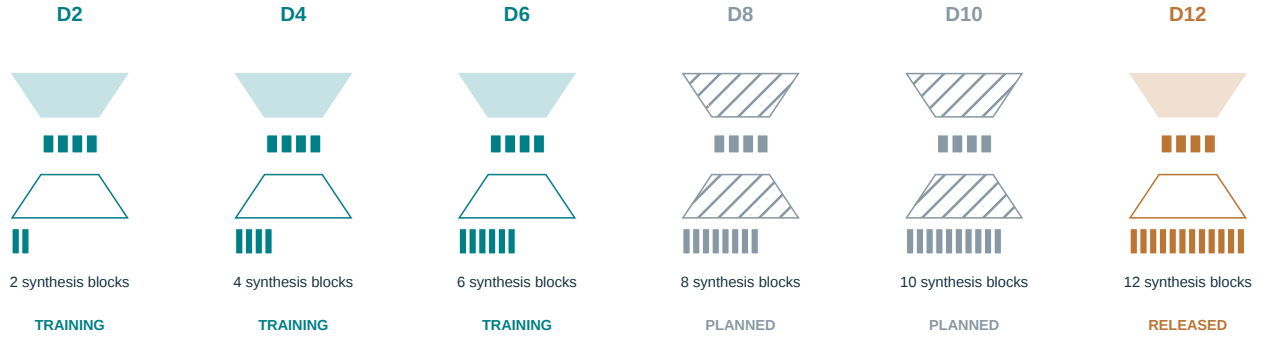
$$\frac{T(d)}{T(12)} = (1 - f) + f \frac{d}{12} + h. \quad (\text{S4})$$

We sweep $f \in \{0.5, 0.7, 0.9\}$ and $h \in \{0, 0.03\}$. These are explicit sensitivity assumptions, not fractions estimated from the hardware. The reciprocal of Eq. (S4) is the plotted scenario speedup. It shows why removing most trunk blocks need not yield a proportional reduction in complete codec time. The curves deliberately contain no predicted PSNR, bitrate or empirical uncertainty bands.

Exact decoder size is 1,767,168 + 1,038,720 d parameters; adding the 27,947,520 non-synthesis parameters gives complete-codec size. This formula matches all three independent training manifests. It describes capacity, not compute: retained tensors, arithmetic and measured device time are separate quantities. The data file labels these scenarios explicitly and records every assumption used to generate them. A separate CPU construction of all six depths verifies these counts and exact equality of their retained initial tensors to the seed-matched D12 initialisation; it performs no training or quality evaluation.

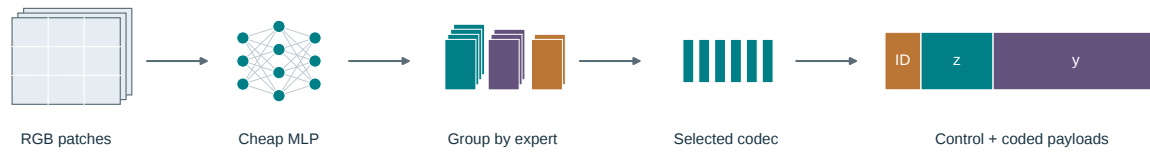
S11. Encoder Information Acquisition

Figure S8 separates candidate reconstruction from error-table construction, control search and the final mixed reconstruction. A single trunk pass can tap all reachable depths, reducing repeated prefix work. The historical timing summary lacks raw repetition uncertainty; it cannot be used as a paired end-to-end benchmark. In addition, the archived GPU search timer may include pending reference work, so it is excluded from the isolated search comparison. The runtime audit in the paper repository also identifies an older neural replay benchmark whose outputs are not causally reconstructed from decoded stream symbols. Its stage labels must not be read as a verified complete codec.

a Independent codecs: depth changes the synthesis network, training changes the whole codec

D2 / D4 / D6: pinned official recipe, all codec weights learned jointly

A same-recipe D12 control is still needed to separate depth from training provenance.

b Proposed deployment: choose before encoding, group work by expert, transmit the choice

The decoder reads the expert ID and uses that expert's entropy model and synthesis transform.

Measure: full bytes · encoder + decoder wall time · final cropped quality · bank memory

Deployment is a proposed design. The glyphs and queues are schematic; they do not show measured routes or throughput.

Figure S6. **An independent codec bank is a separate experimental system.** Training status is shown as of 27 September 2026. D2/D4/D6 are in progress, D8/D10 are planned, and D12 is the released reference. Each expert owns its analysis, entropy and synthesis parameters. The proposed encoder-side MLP selects an expert before encoding; the decoder receives the expert ID. Queues, network shapes and stream segments are schematic. No throughput or quality result is encoded by their geometry.

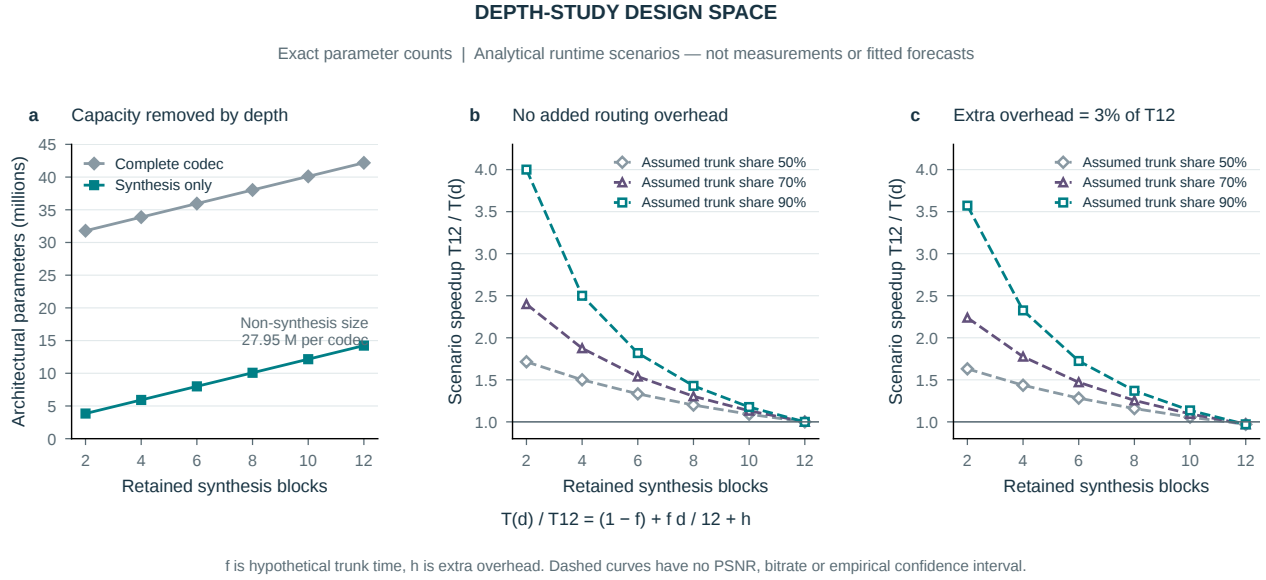
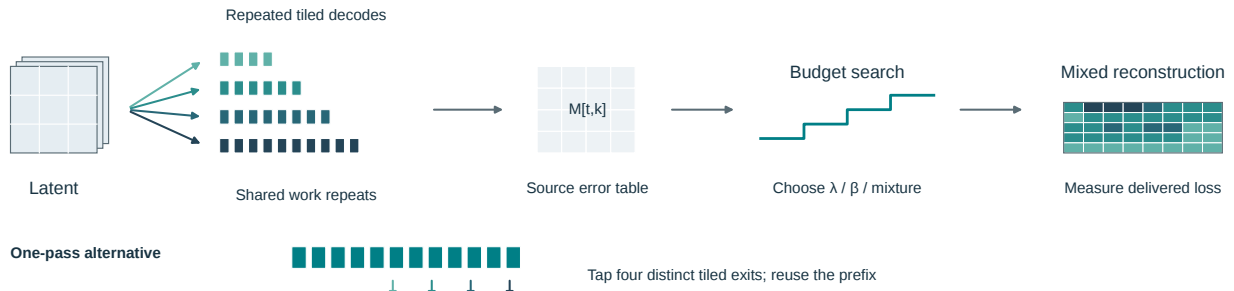
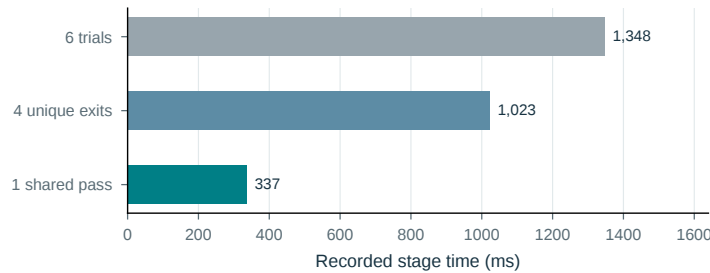


Figure S7. **A transparent design-space illustration for unfinished experiments.** **a**, Exact architectural parameter counts. **b,c**, Analytical runtime sensitivity to assumed trunk-time share and added overhead. Open markers and dashed lines denote assumptions, not measurements or fitted forecasts. No quality or bitrate outcome is assigned to an unfinished model.

a The source is available at the encoder, but the error table must be computed



b Recorded table construction time



Historical timing: 8 frames $\times$ 3 repetitions at QP32. No new end-to-end latency measurement.

c Calibration changes the cost

Fixed, held-out β

Router uses decoded features.
No per-frame source table is required.

Per-frame source-calibrated β

The plotted budget sweep uses $M[t, k]$.
A fast router does not remove that cost.

Figure S8. **The decision path has an encoder cost.** **a**, Stages required for source-error-based selection, with prefix reuse across candidate exits. **b**, Historical table-construction upper medians over 24 pooled trials, without invented uncertainty bars. **c**, A fixed held-out control and per-frame source calibration require different information. This is a stage analysis, not an end-to-end speedup.

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